

## Announcements

- CSTEM Symposium Wednesday, March 25  
9-5  
Athletics Building
  - Quiz on Control Flow w/PHP to be posted tonight. Due before class Tuesday
- 

## Review of last time

**functions** are a group of statements given a name & perform a specific task when called.

**function** keyword used to define a function

```
// simple function example
function display_store_name(){
    echo "Widgets & McGuffins";
}
```

## Getting Data out of a Function

**return** statement returns data from a function to the place where the function was called.

keyword      name      parentheses

```
function get_store_name(){
    return "Widgets & McGuffins";
}
```

returns the value to the place where it is called

```
<p>The store is called <?= get_store_name() ?></p>
```

\$company = get\_store\_name();

`$Company = get-store-name();`

## Getting Data Into a Function

**parameters** are placeholder variables that provide data to a function

Keyword      function name      parameter variable

```
function find_store_name($name){  
    return "The store is $name."  
}
```

parameter variables

```
function display_product($name, $price){  
    echo "<p>$name - \${$price}</p>";  
}
```

"n"  
"n"

Sprocket - \$4.99

`\$` - escapes the \$ so that it prints it.  
arguments passed to param variables

```
display_product("Sprocket", 4.99);
```

**arguments** are values passed to parameter variables when the function is called. Can either be literal values or variables.

```
function calculate_cost($price, $quantity){  
    $cost = $price * $quantity;  
    $tax = $cost * (10/100);  
    return $cost + $tax;  
}
```

`$result = calculate_cost($item_cost, $quantity);`

```
<p>The cost of <?=$quantity?> $<?=$item_cost?> items with  
<?=$tax?>% is <?=$result?></p>
```

The **scope** of a variable is where it can be seen in a program. Params & vars defined inside a function can only be seen inside that function.

## Default Values

Make functions more flexible

```
function calculate_cost($price=5, $quantity=1){  
    $cost = $price * $quantity;  
    $tax = $cost * (10/100); //0.1  
    return $cost + $tax;  
}
```

default value for price

default value for quantity

If no arguments are passed, the default values are used.

```
echo calculate_cost() . "<br>"; ①  
echo calculate_cost(7) . "<br>"; ②  
echo calculate_cost(quantity:20); ③
```

① No params passed @ at all. So both default values used.

② A single value passed. Default parameters are positional, so the value passed here is for price, quantity stays @ 1.

③ To send an argument to a specific param, use the name of the param, followed by a colon & the value.